

The morbidity and outcome of patients with Guillain–Barré syndrome admitted to the intensive care unit

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Abstract

One third of patients with Guillain–Barré syndrome (GBS) require admission to the intensive care unit (ICU), associated with significant risk of morbidity, mortality, and incomplete recovery.

Methods: 76 adult patients with GBS admitted to the ICU at a regional referral center over a 20-year period were studied. We determined the frequency, nature, and predictors of complications they experienced while in the ICU; this morbidity was related to long-term functional recovery and time to regain independent ambulation, extracted from longitudinal follow-up data.

Results: ICU stay was a median 21 days and mechanical ventilation (MV) was required in 78% (median duration 28 days). Two-thirds suffered at least one major complication, most commonly pneumonia (54%). Morbidity was strongly associated with MV and male sex. Mortality occurred in only 5 patients (6.5%). Over an average 3 years follow-up, recovery of independent ambulation was seen in 75%, with advanced age being the most powerful predictor of poor outcome. Prolonged MV and severe axonal loss did not preclude a favorable recovery. Time to ambulate was a median 198 days, although recovery could occur as late as ten years after onset; slower recovery was associated with ICU complications, prolonged MV, and early axonal abnormalities.

Conclusion: Although patients with GBS suffer significant morbidity during protracted ICU stays, with meticulous supportive care, many make gratifying functional recoveries. In severely afflicted patients, this may only be appreciated after extended follow-up.

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1. Introduction

Guillain–Barré syndrome (GBS) manifests as an acutely evolving paralytic illness that may rapidly progress to quadriplegia, weakness of bulbar and respiratory musculature, and autonomic nervous system dysfunction [1]. Approximately one third of GBS patients are admitted to an intensive care unit (ICU); many require mechanical ventilation (MV), often for prolonged periods [2–4]. During this critical phase, they are at risk for systemic complications with the potential for substantial morbidity and consequent mortality [5,6]. These

severely afflicted patients have a less favourable prognosis for functional recovery, suffer greater long-term disability, and expend substantial personal and health-care costs [7]. The nature and extent of morbidity experienced by these patients while in the ICU and its impact on outcome has only been explored in a limited manner [8,9]. This knowledge could help identify correctable factors, optimize medical care, and guide discussions of prognosis with patients and their families.

We undertook a retrospective review of all adult patients with GBS admitted to the ICU at our institution over a twenty year period to delineate the spectrum of complications they suffer while critically ill, to determine the rates of severe morbidity and mortality, and to assess long-term functional outcome. We extracted predictors of complications and of

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eventual recovery, and determined specifically how ICU morbidity impacts on rate of recovery and long-term prognosis.

2. Methods

The medical records of all patients with a diagnosis of GBS who were admitted to the ICU at London Health Sciences Centre between 1983 and 2003 were reviewed. We excluded children and those subsequently found to have mimics of GBS, including CIDP. Patient demographics and medical comorbidities were recorded; organ dysfunction (including cardiac and pulmonary disease), diabetes mellitus, and coronary artery disease were considered serious comorbidities. Respiratory or gastrointestinal infections preceding the onset of symptoms and the temporal evolution of weakness were recorded. Peak clinical severity was quantified by degree of maximal upper limb weakness using the following ordinal scale: 0, normal; 1, weak, but able to lift arms off bed; 2, flicker of movement; 3, no movement. The primary reason for ICU admission, presence of overt bulbar and autonomic dysfunction (cardiac arrhythmias including sinus tachycardia, labile blood pressure, or gastrointestinal dysmotility), need for and duration of MV and performance of tracheostomy were documented. All immunomodulatory treatments received were noted.

Baseline and follow-up electrodiagnostic studies were reviewed and each patient was classified into electrophysiological subtypes based on their initial study as defined by Hadden et al. [10]. We determined whether patients developed low compound muscle action potential (CMAP) amplitudes ($<20\%$ lower limit of normal [LLN] in at least two nerves) or inexcitable nerves (CMAP absent in all nerves or $<10\%$ LLN in one nerve and absent in all others tested) at any point during their illness.

2.1. ICU complications

Records were carefully scrutinized for all ICU complications, including hyponatremia (serum sodium <130), hyperglycemia requiring persistent treatment, and elevations of hepatic transaminases (greater than $2\times$ upper limit of normal). Major morbidity was defined as: serious infection (pneumonia, sepsis), deep vein thrombosis (DVT) or pulmonary embolism (PE), severe arrhythmia (brady- or tachyarrhythmias requiring acute intervention, including cardiac arrest), gastrointestinal haemorrhage, complications of tracheostomy, pseudomembranous colitis due to *Clostridium difficile*, ileus with or without bowel perforation. Pain and requirements for analgesia were noted.

2.2. Outcomes

Outcome measures included: length of stay (LOS) in the ICU and the hospital; mortality in the ICU, at one-year, and at maximal follow-up. Functional recovery was extracted from standardized prospectively collected assessments of motor scores (MRC scale) and gait recorded at each follow-up clinic

visit. To achieve uniformity with similar studies, this data was translated into the *Hughes' scale*: 0, healthy; 1, minor symptoms or signs, able to run; 2, able to walk >5 m without assistance but unable to run; 3, able to walk >5 m with assistance; 4, bed- or chair-bound; 5, requiring assisted ventilation for at least part of the day; 6, dead [11]. Grades were assigned at 6 months and 1 year from disease onset, then from yearly follow-up as available. Good outcome was defined as recovery of independent ambulation (grade ≤ 2) at final follow-up. The distribution of functional status was also charted at each time point to depict serial improvement, assuming those achieving a good outcome continued to walk independently at all subsequent points. Time to walk independently from disease onset was also approximated from follow-up records.

2.3. Statistical analysis

Baseline characteristics and outcomes are presented descriptively. Logistic regression was used to examine the association between the pre-specified variables and occurrence of serious ICU complications, poor functional outcome and time to ambulation. Characteristics significant at the 0.10 level were considered for entry into multivariate analysis. Associations with a p value of less than 0.05 were considered to be statistically significant. Data were analysed using SAS 8.2. The study methodology was approved by the Institutional Review Board.

3. Results

Demographic information and the clinical characteristics of the 76 patients eligible for study are shown in Table 1. The majority of these GBS patients experienced a fulminant disease course, requiring ICU admission a median of 5 days from onset; peak severity was reached after a median of 8 days with 74% of patients having complete paralysis or only a flicker of upper limb movement. Indication for ICU admission was respiratory failure in most cases; most patients with bulbar dysfunction (87%) required intubation. Almost all patients received immunomodulatory therapy.

3.1. Electrophysiology

A significant proportion (19%) were classified as having axonal disease based on initial testing, while 20% overall developed severely reduced motor amplitudes and 11% developed electrically inexcitable nerves during the course of their illness. Axonal abnormalities on initial study were seen more frequently in patients with a preceding diarrheal illness than in those without (31% vs. 15%), but this did not reach statistical significance. Initial classification was strongly associated with peak motor weakness: 64% of patients whose first EMG studies were classified as axonal or inexcitable progressed to complete quadriplegia, compared to only 24% of those classified otherwise (OR 5.6, 95% CI

Table 1
Clinical features of the study population

Characteristic	Frequency (%) (<i>n</i> =76)
Age, years (mean±SD)	56.1±16.9
Median (range)	58 (19–87)
Age>60 years	32 (42)
Male sex	46 (61)
Significant medical comorbidity	12 (16)
Preceding infection	52 (68)
Respiratory	40 (53)
Gastrointestinal	16 (21)
Days to hospital admission (mean±SD)	3.1±3.9
Days to ICU admission (mean±SD)	6.8±6.6
Days to peak severity (mean±SD)	9.9±6.3
Peak upper limb weakness	
Normal (Grade 0)	1 (1)
Antigravity (Grade 1)	19 (25)
Flicker (Grade 2)	32 (42)
Complete paralysis (Grade 3)	24 (32)
Reason for ICU admission	
Respiratory failure	54 (71)
Airway protection	12 (16)
Observation	8 (11)
Autonomic instability/arrhythmia	2 (3)
Bulbar dysfunction	60 (79)
Autonomic dysfunction	53 (70)
Mechanical ventilation	59 (78)
Duration of MV, days (mean±SD)	45.2±54
(median, range)	28 (2–329)
MV>14 days	44 (75) ^a
MV>60 days	16 (27) ^a
Tracheostomy	33 (56) ^a
Immunomodulatory treatment	
Plasmapheresis (PLEX)	23 (30)
Intravenous immunoglobulin (IVIG)	37 (49)
PLEX+IVIG	5 (7)
PLEX+steroids ^b	5 (7)
None ^b	6 (8)
Retreatment given ^c	15 (20)
Initial electrophysiological classification	
Normal	3 (4)
Equivocal	9 (12)
Demyelinating	50 (66)
Axonal	12 (16)
Inexcitable	2 (3)
Low compound muscle action potentials	15 (20)
Inexcitable nerves (at any point)	8 (11)
Absent sensory nerve action potentials	18 (24)

^a Percentage of those requiring MV.

^b Steroids or no treatment (placebo) were only given in the context of early randomized trials in the 1980's.

^c Second course usually given for recurrent or progressive clinical weakness.

1.6–19.4; $P=0.008$). Quadriplegic patients were more likely to develop low CMAP amplitudes (OR 29.6, 95% CI 5.8–150.1; $P<.001$) or inexcitable nerves. This most severely affected group also needed MV more frequently (96% vs. 61%, $P=0.03$) and for a longer duration (75 vs. 17 days, $P<.001$); their disease progression was more fulminant, with a shorter mean time to hospital admission (1.6 vs. 3.7 days, $P=0.008$), to ICU admission (3.8 vs. 8.2 days, $P=0.007$),

and to peak motor severity (8.0 vs. 10.7 days, $P=0.10$). Prolonged ventilatory support (longer than 14 days) was more likely required by patients in whom initial electrophysiological test results met axonal criteria (86% vs. 52%, $P=0.03$); all those who developed inexcitable nerves required MV for 60 days or longer.

3.2. ICU complications

Two-thirds of GBS patients suffered at least one serious complication during their stay in the ICU (Table 2); one-quarter had three or more serious complications. Cardiac dysrhythmias were observed most commonly during the early disease course and were associated with signs of generalized autonomic instability. They manifested as supraventricular tachycardia, severe bradycardia and complete heart block (requiring insertion of a temporary pacemaker in three patients). Six patients suffered cardiac arrest associated with either asystole (often preceded by bradycardia) or ventricular fibrillation, the latter being fatal in two of three cases. These severe cardiac events occurred within the first month of ICU stay, often in association with other critical complications.

Additional morbidity resulted from: urinary tract infections in 43 (57%), hyponatremia in 35 (46%), elevation of liver enzymes in 20 (26%), and hyperglycemia requiring treatment in 10 (13%). Three patients suffered a pneumothorax; one with central line insertion for plasmapheresis (PLEX). Pain was almost universal and usually severe, most often reported as back and radicular pain at disease onset, followed by musculoskeletal deep aching pain while immobile in the ICU, and burning dysesthetic pain and allodynia later in the course; most patients were treated with opioids, often combined with tricyclic antidepressants, anticonvulsant drugs such as gabapentin, and NSAIDs.

The rate of complications did not change over the two decades of study. Age, comorbidity, rate of disease progression, preceding illness, initial electrophysiological classification and immunomodulatory treatment received were not associated with risk of ICU complications. However, there was a strong association with MV (Table 3); specifically, complications occurred in 15 of the 16 patients ventilated for more than 60 days. The higher rate of morbidity observed in

Table 2
Serious ICU complications

Complication	Frequency (%)
Pneumonia	41 (54)
Sepsis	18 (24)
Severe dysrhythmias	17 (22)
Ileus and/or bowel perforation	13 (17)
Deep venous thrombosis ^a	7 (9)
Pulmonary embolism ^a	5 (7)
Gastrointestinal hemorrhage	5 (7)
Pseudomembranous colitis	5 (7)
Complications of tracheostomy	3 (4)
At least one serious complication	51 (67)

^a Either a DVT and/or PE was suffered by a total of 10 patients (13%).

Table 3
Factors associated with serious ICU complications

Variable, n (%)	Complications (n=51)	No complications (n=25)	P value	Odds ratio (95% CI)	Multivariate OR (95% CI)
Male sex	35 (68)	11 (44)	0.05	2.8 (1.04–7.5)	3.9 (1.02–14.5)
Max arm weakness: grade 2–3	45 (80)	11 (44)	<.001	9.5 (3.0–30.5)	
Max arm weakness: grade 3	21 (41)	3 (12)	0.02	5.1 (1.4–19.4)	
Bulbar involvement	45 (88)	15 (60)	0.005	5.0 (1.6–16.1)	
Low CMAP amplitude	14 (28)	1 (4)	0.01	9.1 (1.1–71.4)	
Absent SNAPs	16 (31)	2 (8)	0.04	5.3 (1.1–25)	
Inexcitable nerves	8 (16)	0	0.05	N/A	
Mechanical ventilation (MV)	49 (96)	10 (40)	<.001	37.0 (7.2–200)	43.5 (7.8–250.0)
Duration of MV (mean±SD)	48.5±57.1	7.8±15.1	<.001	1.08 (1.04–1.13)	
MV>14 days	39 (76)	5 (20)	<.001	13.0 (4.0–41.7)	
MV>60 days	15 (29)	1 (4)	0.03	9.6 (1.2–77.5)	

males was largely due to a twofold greater incidence of pneumonia (67% vs. 33%; $P=0.005$). Pneumonia was also highly associated with duration of MV, especially when prolonged more than 14 days (OR 5.2, 95% CI 1.9–14.1, $P=0.001$). Sepsis developed solely in ventilated patients, with increasing risk for those with prolonged MV (OR 19.6, 95% CI 2.4–167.0 for MV >14 days, $P=0.005$). The risk was highest (75%) for patients with inexcitable nerves, who also all developed pneumonia while in the ICU.

3.3. Outcomes

Mean ICU LOS was 41 days (median 21 days, range 6–335). Hospital LOS at our institution averaged 71 days (median 49 days); only 9 (13%) patients were discharged home, the remainder returned to the referring hospital or a rehabilitation facility. GBS patients suffering serious complications while in the ICU had significantly longer ICU and hospital stays (median of 36 vs. 5 ICU days, Hazard Ratio 4.8, 2.7–8.2; $P<.001$, and 74 vs. 25 hospital days, HR 4.3, 2.5–7.7; $P<.001$) and were much less likely to be discharged home (6% vs. 29%; $P=0.025$).

Mortality in this cohort was seen in 4 patients (5%) while in the ICU, and 5 (6.5%) at both 1-year and maximal follow-up. No deaths occurred in patients with mild weakness or in those not requiring MV; mortality was still only 8.5% amongst ventilated patients. As a whole, mortality risk was not significantly increased in patients experiencing complications (8% vs. 4%; $P>.999$). However, ileus and bowel perforation were often diagnosed late and associated with a high risk of death (23%). The strongest predictors of mortality were increasing age (OR 1.08 per year) and significant comorbid illness (33.3% if present vs. 1.6% if absent; OR 31.5, 3.1–317.8, $P=0.002$). The median age of those who died was 72 years compared with 58 years for survivors. All deaths occurred in ventilated patients who had suffered serious ICU complications.

Extended follow-up data was available over a mean of 1120 days from disease onset (median 433 days, range 16–6931). Of those surviving, 49 patients (70%) had at least one year follow-up; 35 (50%) had follow-up for at least

two years, 30 (43%) at least three years, and 26 (37%) were followed for four years or more. At their final follow-up visit, 56 patients (75%) had achieved a good outcome and were able to walk independently (Fig. 1). However, almost one-third of our patients with severe GBS followed long-term (14 of 49) had not recovered independent ambulation by one year, the end-point of most previously reported studies. Ten of these patients (18%) recovered functional ambulation more than a year after disease onset; thus, their outcome would have been erroneously classified as poor at one year. With longer follow-up the majority of our patients eventually attained a good functional recovery, with only 3 of 26 (12%) followed for at least 4 years still unable to walk. This gradual progressive improvement in functional status is depicted in Fig. 2.

Outcome was analysed in relation to major predictive variables (Table 4). Importantly, presence of severe axonal degeneration and complete quadriplegia, although strongly predicting prolonged MV and ICU stays, did not preclude an eventual good recovery. Six of eight patients with completely inexcitable nerves ultimately recovered independent ambulation, although they required ankle-foot orthoses to walk securely due to residual distal weakness. Overall, 42 of 59 (71%) ventilated patients recovered independent ambulation compared to 14 of 16 among those not requiring MV (88%,

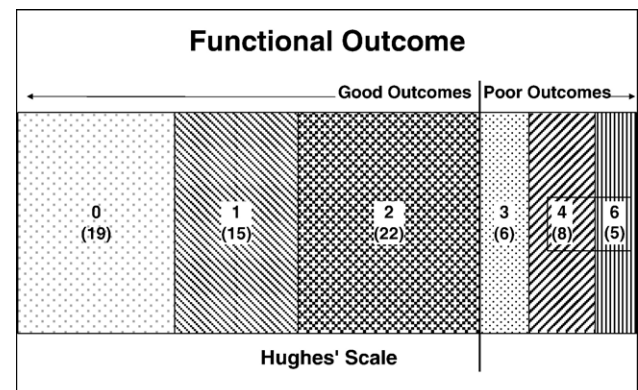


Fig. 1. Functional outcome at last follow-up.

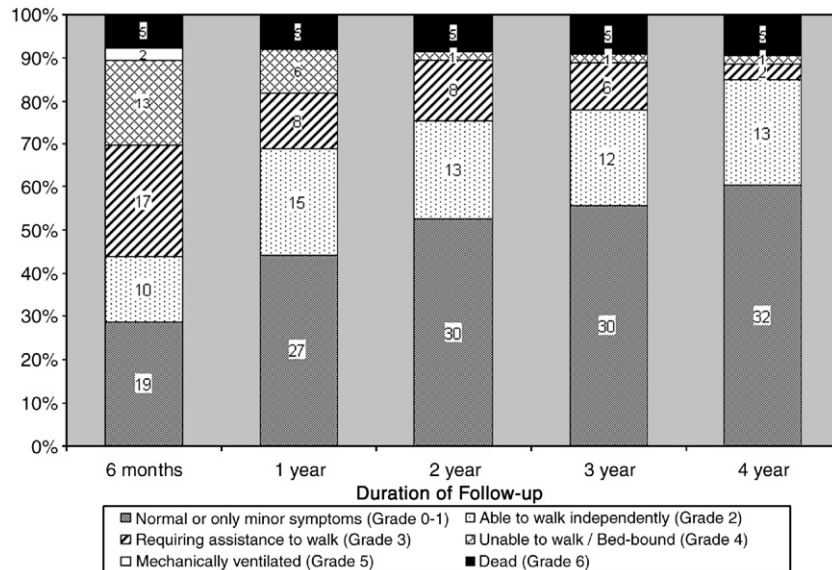


Fig. 2. Functional recovery with serial follow-up.

$P=0.33$). Even 11 of 16 patients who had been ventilated for more than 60 days eventually had a good final outcome.

Patients with more fulminant disease course, i.e. those with a shorter time from symptom onset to hospital admission, tended to have a worse outcome. However, in the final multivariate analysis, only age and medical comorbidity were significantly associated with poor outcome. Although occurrence of serious complications appeared to portend a worse prognosis, the risk difference did not attain statistical significance.

Outcome between groups receiving various immunomodulatory treatments was not statistically different. A trend to shorter duration of MV was seen with implementation of specific therapies (IVIG 33.1 days, PLEX 52.2 days vs. supportive care 113.2 days, $P=NS$). Over the two decades of study, there was no trend seen towards improved rates of recovery.

3.4. Time to ambulation

Estimated time to recover independent ambulation was a median 198 days (range 9–3983 days). Factors associated with

longer time to recovery are shown in Table 5 and include clinical and electrophysiologic markers of disease severity, duration of MV, and occurrence of ICU complications. Rates of recovery are contrasted in Fig. 3 a and b dividing subjects by variables most strongly predicting longer recovery times (serious ICU complications, MV longer than 14 days). In general, recovery was slow in the most severely affected patients, who were quadriplegic with severe axonal degeneration; they endured protracted ICU stays while ventilated and suffered serious ICU complications. Nonetheless they could make substantial functional improvements many years after symptom onset.

4. Discussion

This study examines the subgroup of patients with the most severe GBS, those requiring admission to the ICU, usually for impending respiratory failure, bulbar weakness resulting in inability to protect their airway, or autonomic instability. It

Table 4
Factors associated with functional outcome

Variable, n (%)	Good outcome (n=56)	Poor outcome (n=19)	P value	Odds ratio (95% CI)	Multivariate OR (95% CI)
Age, years (mean±SD)	52.2±16.9	67.6±11.7	0.002	1.08 (1.03–1.13)	1.08 (1.03–1.13)
Significant comorbidity	5 (9)	7 (37)	0.008	5.9 (1.6–22.0)	
Days to admission (mean±SD)	3.5±4.4	1.8±1.6	0.06		
Serious ICU complications	35 (63)	16 (84)	0.09	3.2 (0.8–12.3)	

Table 5
Factors associated with time to recovery

Variable	Hazard ratio (95% CI)	P value	Multivariate HR (95% CI)
Age>60 years	1.5 (0.87–2.7)	NS	
Preceding GI illness	2.0 (0.99–4.0)	0.05	
Initial EMG: axonal or inexcitable	2.1 (1.1–4.2)	0.04	2.9 (1.4–6.0)
Max arm weakness: grade 2–3	2.4 (1.3–4.4)	0.004	
Max arm weakness: grade 3	2.6 (1.4–4.7)	0.002	
Mechanical ventilation (MV)	2.4 (1.3–4.4)	0.006	
MV>14 days	3.2 (1.8–5.5)	<.0001	2.5 (1.4–4.8)
MV>60 days	2.6 (1.3–5.0)	0.005	
Low CMAP amplitude	2.6 (1.3–5.1)	0.007	
Immunomodulatory retreatment	2.1 (1.1–4.4)	0.04	
Serious ICU complication	3.2 (1.8–5.7)	<.0001	2.8 (1.5–5.4)

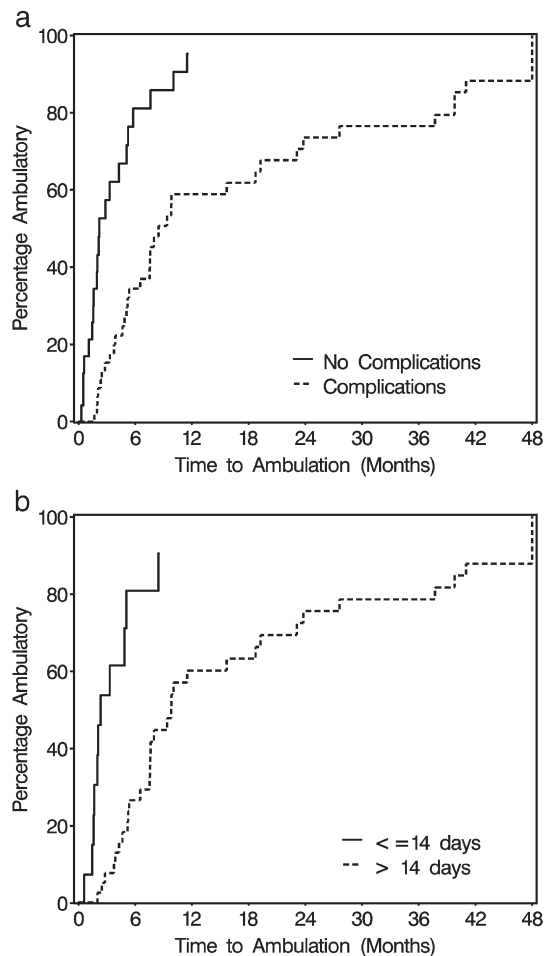


Fig. 3. a: Time to independent ambulation in those with and without serious ICU complications b: Time to independent ambulation in those requiring prolonged mechanical ventilation compared to those not.

confirms that this population has prolonged ICU stays, often coupled with a number of potentially life-threatening complications. The high rate of major complications in our cohort is comparable to that reported in the only other large study from the Mayo Clinic [8]. Yet, a significantly greater proportion of our patients required MV, the factor we found most strongly associated with ICU morbidity.

The spectrum of complications encountered was not unexpected. They encompassed the typical sequelae of prolonged immobility and critical illness. Infections were most common, including pneumonia seen in over half our patients and symptomatic bacteremia observed in almost a quarter. These observations underline the importance of measures to minimize nosocomial infections [12]. Pneumonia, strongly associated with duration of MV, was observed with similar frequency in a study from Queen Square and was found in the majority (78%) of ventilated patients in another recent report [9,13]. Early tracheostomy in patients destined for prolonged MV may be one means of reducing the incidence of this serious complication [14].

Despite almost ubiquitous utilization of heparin prophylaxis in our patients, venous thromboembolism posed a

substantial risk and should be anticipated in these often quadriplegic and critically ill patients who remain immobilized for long periods. Gastrointestinal tract dysmotility is a further hazard of prolonged immobility, compounded by autonomic dysfunction and the frequent need for narcotic analgesia in these patients. This manifests first as constipation and then obstipation progressing to ileus, a derangement found in 15% of severely affected GBS patients in another series [15]. We found ileus, associated with dilatation of the cecum (Ogilvie syndrome) and risk of bowel perforation, in 17% of our population; it was the single complication most strongly associated with mortality. Therefore, use of an aggressive bowel regimen coupled with close monitoring of gut activity should be a cornerstone of ICU management in these high-risk patients [16]. As was demonstrated in a previous prospective study, pain is very common in GBS, and poses significant challenges for recognition and management [17]. It is often multifaceted with musculoskeletal and neuropathic components that interact and evolve as the disease progresses. Use of opioids can often not be avoided, but should be combined proactively with other medications of demonstrated efficacy in this situation [18].

The ability to artificially ventilate patients with respiratory failure for prolonged periods has reduced the overall mortality of GBS [6]. However, a significant risk of death still remains, attributable to the complications of critical illness outlined above. Advances in supportive care, such as improved thromboembolism prophylaxis and careful cardiac monitoring may further mitigate these risks [16]. Nonetheless, a contemporary study from the Mayo Clinic encountered a mortality of 20% in ventilated GBS patients [7]. As confirmed in our study, mortality was greater in the elderly and in those with comorbid illnesses. Of 76 patients in our series, only 4 died during their ICU stay; deaths were due to lethal arrhythmias, fulminant infections, or intestinal perforation, and occurred in patients with severe weakness requiring ventilatory support. There was only one additional death outside the ICU, which occurred after hospital discharge and was unrelated to GBS or any ICU complication. The mortality rate of 8.5% amongst our group of ventilated patients attests to the value of vigilant supportive care aimed at preventing fatal ICU complications. Even though compared to other studies our fatality rate is low for this high risk population, we remain challenged to further reduce this figure by minimizing ICU morbidity.

Predictors of ICU complications conformed to typical indicators of GBS severity, including peak arm weakness, a simple clinical measure, as well as electrophysiological markers of axonal degeneration. Morbidity was far greater in ventilated patients; it increased with longer duration of MV, and was associated with longer ICU LOS. These findings echo those from the study at the Mayo Clinic [8]. GBS patients suffering serious complications tended to have worse outcomes and endured significantly longer times to eventual recovery.

We did not find any particular treatment modality, including PLEX and corticosteroids, associated with an increased risk of

complications (although only a few patients received corticosteroids in conjunction with an early treatment trial). Our series spans the time when immunomodulatory therapies such as PLEX and IVIG became standards of care. Thus the numbers of GBS patients treated with only supportive care are too small to allow comparative analysis with regards to shortening the duration of MV or ICU LOS. The same holds true for the comparisons of various treatments which failed to show significant difference. Previous controlled trials have demonstrated significant reductions in duration of MV with active immunomodulatory therapy [19].

A particular strength of our study was the extended follow-up available for many patients, with serial quantitative functional assessments routinely collected; these were converted into the Hughes' scale as an indicator of functional recovery. We did not limit the evaluation of recovery to an arbitrary duration of one year, but followed a significant proportion of patients for many years. Over this prolonged period of observation, many improved gradually to ultimately make gratifying recoveries, thereby validating months of ICU care and years of structured rehabilitative efforts. Abbreviated follow-up clearly underestimates this potential for slow and delayed recovery among these severely affected GBS patients who are often still ventilated and bed-bound six months to one year after disease onset [5]. The group from Queen Square illustrated this important lesson in describing a patient who had been ventilated for 910 days but who still eventually recovered independent ambulation [9].

Patients classified as having early features of axonal disease according to previously published criteria [10], had a longer duration of MV and required a significantly longer time to recover independent ambulation. Some patients with GBS who meet criteria for primary axonal disease may recover rapidly when the immune-mediated damage is restricted primarily to intramuscular motor nerve terminals, the most distal nerve segments, or results in immune-mediated physiological block at nodes of Ranvier [20]. Our findings suggest that in general the group of patients with primary axonal changes on initial electrophysiological study had a distinctly worse prognosis compared to those with changes of primary demyelination, with few making such rapid recoveries. Late axonal degeneration, manifesting as progression to low CMAP amplitudes and inexcitable nerves, while often secondary to severe inflammatory demyelination, also predicted prolonged MV and ICU stay with its inherent risk for serious complications, and a slow recovery.

A good outcome, defined as the ability to walk independently, was seen in 75% of our patients (including 71% of those requiring MV). This is comparable to findings of the Mayo Clinic cohort [7]. Among their ventilated patients, a good outcome was seen in 61% and improvement beyond one year was observed in 19% of those followed for longer, a figure almost identical to our 18% rate for delayed improvement. Many previous studies had suggested that long duration of MV was uniformly associated with poor outcome [5,7]. Our findings do not concur with this statement; while we

found that advanced age and presence of medical comorbidity did portend a poor prognosis, even those patients with inexcitable nerves who were ventilated for many months often attained a favourable outcome, despite a significantly slower trajectory to recovery. This caveat was voiced in another study where duration of MV was associated with short-term outcome (at 3-months and even 1-year) but not with long-term recovery recorded 3 years from disease onset [9].

These observations make a strong case for continued long-term supportive care for even the most helplessly quadriplegic GBS patients enduring months of assisted ventilation and exhibiting signs of extensive axonal degeneration. If serious complications like nosocomial infections and ileus can be avoided while in the ICU, the prolonged disability incurred by this critical illness can often be overcome with sufficient time and rehabilitation. More cautious optimism may be exercised in those who are elderly or suffer from a serious underlying illness, whose chances of good recovery are significantly lower.

The limitations of this study and subsequent analysis are inherent in its retrospective nature. Details of complications were extracted from medical records and the analysis of patient outcome was based on available patient follow-up examinations. Retrospective grading of clinical outcome and especially timing of functional improvement can only be considered estimates. The variables found to be related to important outcomes, such as morbidity, mortality, and recovery, are not necessarily causal in their relationship. These associations may be linked in either direction, or only be related by other unseen confounders. However, the plausibility of the findings, and consistency with other similar studies, makes this less likely. We feel that this is a useful descriptive analysis of a complex disease and its protracted course. Understanding the full-spectrum of severe GBS will stimulate enhanced patient care, rational prognostication, and ultimately improve outcome.

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